



Chapter Notes: Adolescence: A Stage of Growth and Change

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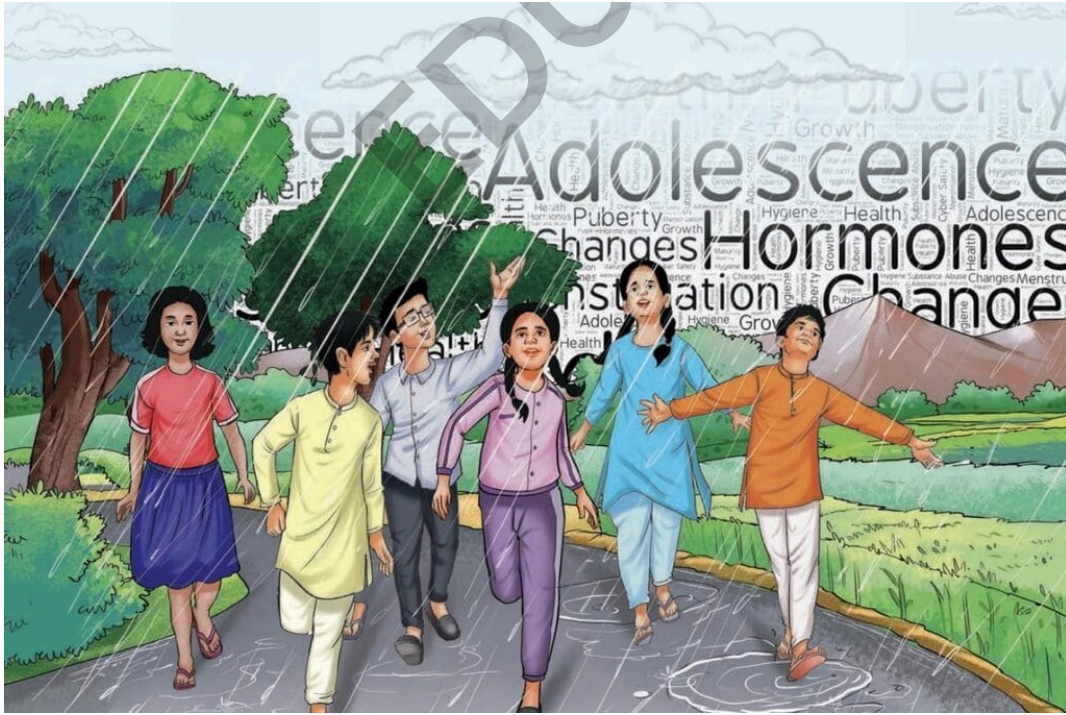
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Introduction

The life of a plant starts from a tiny seed. With proper care, it grows into a sapling and then a mature plant, which can produce flowers, fruits, and new seeds. But it takes time for a plant to grow before it can make seeds again.

Similarly, **animals and humans** also grow and change before they can reproduce. Some animals lay eggs, while others give birth to young ones. These young ones grow slowly and develop over time.

In **humans**, life moves through different stages: **infancy, childhood, adolescence, adulthood, and old age**. Between ages **10 to 19**, we go through **adolescence**, a time of fast physical, emotional, and mental growth. During this stage, the body begins to prepare for adulthood and reproduction.



Let's explore the changes that happen during adolescence and learn how to handle them responsibly.

Growing with Age: The Teenage Years

What is Adolescence?

Adolescence is the transitional stage between childhood and adulthood, starting around age 10 and usually lasting until age 19. It is characterised by rapid physical, emotional and behavioural changes as the body and mind develop toward adulthood.



Changes During Adolescence

As children grow into teenagers, their bodies and minds go through many **physical and emotional changes**. These changes are natural and happen at different times for different people. Here are some common changes:

1. Increase in Height and Body Shape

Boys and girls grow taller quickly during the adolescent growth spurt.

Boys may gain weight, develop broader shoulders and a wider chest.

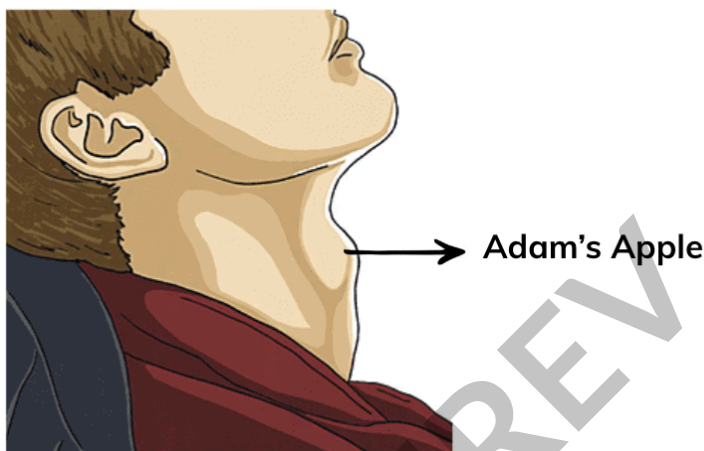
Girls may notice breast development and changes in body shape (wider hips).

2. Change in Voice

Boys' voices become deeper as the voice box (larynx) grows and the vocal cords lengthen.

The visible bump on the throat is called the **Adam's apple**.

Girls may also experience slight changes in voice pitch or tone during adolescence.



3. Hair Growth

Hair begins to grow in the **armpits** and the **pubic area** in both boys and girls.

Boys may develop facial hair such as a moustache or beard and sometimes hair on the chest or back.

The amount and timing of hair growth vary from person to person and are influenced by heredity and hormones.

4. Changes in Skin

Many adolescents get **pimples or acne** because the skin becomes oilier during this time. Acne happens when oil and dead skin cells block pores; good hygiene and, if needed, medical advice can help manage it.



A Key Point to Remember

Everyone goes through these changes at their **own pace**.

Some may grow faster or slower, and that is **completely normal**.

These changes are part of growing up and becoming independent.

Physical Changes

An infographic titled 'Physical Changes' showing four adolescents (two females and two males) standing in a row. To their left are two boxes listing changes for females and males. To their right is a box listing changes common to both. A large, faint watermark 'EDUREV' is visible across the image.

In females:	In both:
- Breasts develop - Waistline narrows - Hips widen - Body fat increases - Menstruation starts	- Body hair appears - All permanent teeth grow in - Perspiration increases
In males:	
- Facial hair appears - Voice deepens - Shoulders broaden - Muscles develop - Hairline begins to recede	

Secondary Sexual Characteristics

Certain changes-such as voice deepening in boys, facial and chest hair growth in boys, and breast development in girls-are called **secondary sexual characteristics**. These characteristics distinguish males from females but are not directly involved in reproduction. They signal the onset of **puberty**, the stage when the body prepares for adulthood and reproductive capability.

Secondary sexual characteristics are physical changes that develop at puberty that help to distinguish sexes, but are not necessary for reproduction.



Secondary sexual characteristics usually appear during puberty and continue to develop until adulthood.

They are caused by hormones produced during adolescence.

What is Puberty?

Puberty is the phase when both external and internal changes prepare the body for **reproduction**. It marks the beginning of adolescence and the path toward becoming an adult.

Internal changes include the development of sex organs and the start of processes necessary for reproduction.

MULTIPLE CHOICE QUESTION

Try yourself: What is the age range for adolescence?

A 10 to 19 years

B 5 to 10 years

C 20 to 25 years

D 1 to 5 years

[View Solution](#)

Changes that Indicate Reproductive Capability

Adolescence brings not just visible changes like height or voice, but also **important internal changes**, especially those related to **reproduction**.

Menstrual Cycle in Girls

The menstrual cycle, commonly called "the period", is a natural process in adolescent girls and a sign of reproductive maturity.

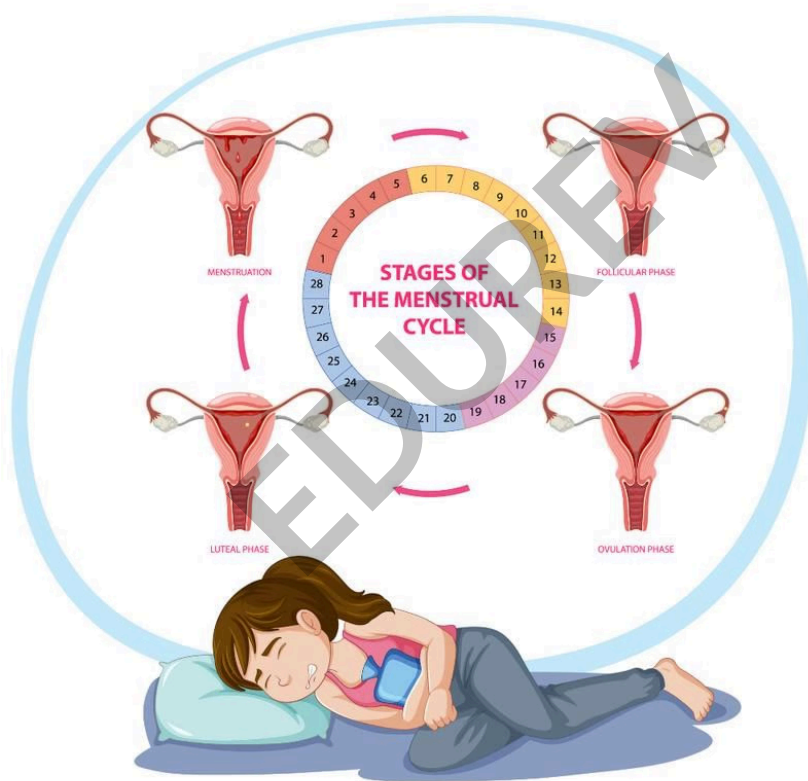
It usually occurs every **28-30 days**, but it can vary between **21-35 days**.

The time when blood flows out from the body is called **menstruation** and it usually lasts for **3 to 7 days**.

Some girls may feel pain or discomfort in the lower belly (menstrual cramps) during this time.

Menstruation is a **normal and healthy** sign of reproductive growth.

Menstruation naturally stops between the ages of about **45-55**, which is called **menopause**.



Menstrual Cycle

Breaking Myths About Menstruation

Many wrong beliefs and taboos exist about periods in different communities; these are not scientifically correct.

Some people believe girls should stay isolated during menstruation-**this is not correct.**

Menstruation is natural and healthy; there is no need for shame or fear.

Understanding periods with a **scientific outlook** supports **women's health** and helps remove unnecessary taboos.

Emotional and Behavioural Changes in Adolescents

Adolescence is not only a time of physical growth - it is also a time of **strong emotions** and **changes in behaviour**. These changes can feel **exciting, confusing, or overwhelming**, but they are **a normal part of growing up**.



Let's understand some common emotional changes and how they affect behaviour, along with ways to handle them positively.

Key Emotional Changes	Probable Effects on Behaviour	Ways for Positive Growth
Mood swings	Feeling happy one moment and upset the next	Participate in music, dance, sports, or pursue hobbies to uplift mood
Strong emotions	Getting angry or sad quickly; feeling stressed	Talk to trusted adults, maintain a journal, or try mindfulness techniques
Increased sensitivity	Feeling easily hurt or misunderstood	Express through art, writing, or explore new interests
Compassion	Feeling concern for others and wanting to help	Engage in social work or community service activities
Curiosity and creativity	Wanting to try new things or take risks	Use creative outlets like reading, painting, or conducting science projects

Understanding and Managing Emotions

Adolescents often feel emotions more strongly than children; mood swings are common. These emotions can lead to new interests such as helping others, joining clubs, or exploring creative hobbies.

By **understanding your feelings** and talking to trusted adults or friends, you can **make thoughtful choices** and respond in a healthy and balanced way.

MULTIPLE CHOICE QUESTION

Try yourself: What is the typical duration of menstruation?

A 3-7 days

B 1-2 days

C 10-14 days

D 5-10 days

[View Solution](#)

Making Adolescence a Joyful Experience

Adolescence is a special journey filled with **curiosity**, **energy**, and **new experiences**. To make this stage happy and healthy, it is important to take care of your body and mind. Good habits, sensible choices and staying positive can make a big difference!

Meeting Nutritional Needs

Adolescence requires a nutritious diet to support rapid growth and development.



Nutrients Needed

Proteins and carbohydrates: Essential for growth, strength and energy; found in foods like milk, pulses, millets, eggs and curd.

Calcium: Supports bone growth and strength; found in milk, cheese, paneer and leafy greens.

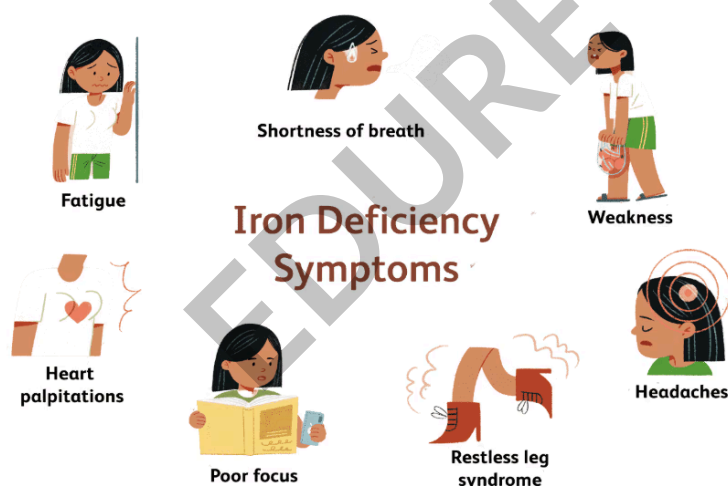
Iron: Aids formation of healthy blood and prevents tiredness; found in spinach, legumes, jaggery and dried fruits like raisins.

Fats, vitamins and minerals: Needed in adequate amounts for overall health; include fruits, vegetables, nuts and oils in balanced amounts.

Health concerns: Adolescents-especially girls-may face iron or vitamin B12 deficiency leading to weakness or anaemia; a balanced diet and timely medical advice help prevent and treat these problems.

Science and Society: Iron Deficiency in Adolescents

Many adolescents, particularly **girls**, may suffer from **iron deficiency**. This can lead to **tiredness**, **weakness** or **anaemia**. To prevent this, eat iron-rich foods, combine them with vitamin C rich foods for better absorption, and follow school health programmes.



Eat **iron-rich foods** such as spinach, lentils, jaggery and dried fruits.

Include **vitamin C** (for example oranges or amla) with iron-rich meals to help the body absorb iron better.

Government programmes like **Iron and Folic Acid (IFA) supplementation** provide tablets in many schools to help prevent deficiency.

Know a Scientist: Dorothy Hodgkin



Dorothy Hodgkin was a noted chemist who studied the structure of molecules including **vitamin B₁₂**, which is essential for healthy blood and nerves.

She received the **Nobel Prize in Chemistry in 1964** for her work in X-ray crystallography.

Our body cannot make vitamin B₁₂, so it must be obtained from food such as dairy, eggs and meat; speak to a teacher or family member to learn more about foods rich in B₁₂.

Personal Hygiene

Along with healthy food, good **personal hygiene** is essential during adolescence.

Because the body goes through many changes, keeping the body clean is important.

Clean the **armpits** and **pubic area** regularly to prevent infections.

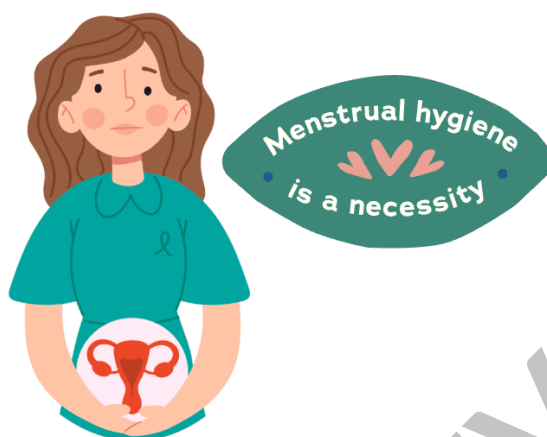
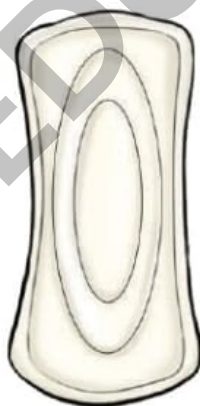
Menstrual Hygiene for Girls

Girls should take extra care of hygiene during **menstruation (periods)**.

Use **sanitary pads** or properly cleaned **reusable cloth pads** to stay clean and comfortable.

Dispose of used pads properly: wrap them in newspaper and throw them in a **covered dustbin**.

Biodegradable sanitary pads are available and are better for the environment.



Breaking the Stigma

Menstruation is **natural** and should not be a cause for shame. Society should:

Provide clean toilets and safe spaces in schools and public places.

Make sanitary products easily available and affordable.

Talk openly about **menstrual hygiene** to support girls and women.

Scheme Name	Purpose
Menstrual Hygiene Scheme (MHS)	Free or low-cost sanitary pads and awareness sessions for girls in rural areas.
Rashtriya Kishor Swasthya Karyakram (RKSK)	Promotes overall adolescent health, including menstrual hygiene through peer education.
Suvidha Sanitary Napkin Initiative	Affordable biodegradable pads sold at Jan Aushadhi Kendras.
State Schemes	Many states like Karnataka (Shuchi), Tamil Nadu, and Odisha run free pad distribution programs in government schools.

Physical Activities & Social Life

Benefits: Regular exercise, games and sports keep the body and mind fit, build stamina and improve mood.

Examples: Activities like running, cycling, swimming or team sports help overall health and social skills.

Importance: Physical activity supports growth, reduces stress and enhances well-being during adolescence.



Balanced Social Life

We all live together in society and interact with many people every day. Being **polite** and **respectful** helps create a friendly and safe environment for everyone.

Adolescence and Social Interaction

Adolescence brings new feelings and experiences; teens may form closer friendships and start to feel romantic attraction.

Many teens copy behaviour of friends as they try to find their own identity.

Many adolescents now interact **online** through social media and messaging apps; these can help learning and friendships but also have risks.

Using Technology Responsibly

Technology helps us connect, learn and share information easily.

Use social media carefully; think before posting and protect your privacy.

Sometimes people may use social media carelessly or hurt others without meaning to.

If you face **cyberbullying** (mean messages, rumours, or sharing private information), tell your parents or teachers right away and save evidence if possible.

Dos	Don'ts
Be respectful and kind	Do not share personal photos with strangers or online friends
Think before posting	Avoid posting harmful or false information
Protect your privacy	Do not share private details with unknown people

Science and Society: Cyberbullying - Cyberbullying means using phones or computers to hurt others by sending mean messages or sharing private information without permission. It is important to stay safe online and seek help if you face it.

Avoiding Harmful Substances

During adolescence, some people may try to pressure you to use harmful substances like **tobacco, cigarettes, alcohol, or illegal drugs**. Because of curiosity and excitement, teens might be tempted to try them.



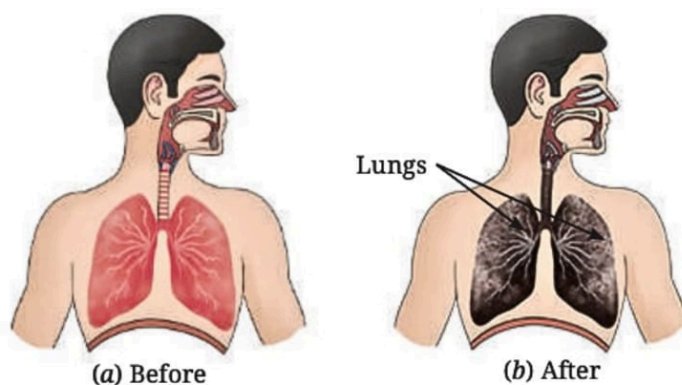
Why Avoid These Substances?

They are **harmful to your body and mind**.

They are **addictive**, meaning once you start you might want to keep using them.

This addiction is called **substance abuse**.

Using these substances can cause serious health problems like **lung damage, breathing difficulties** and **memory loss**.



Condition of the lungs before and after prolonged exposure to bidi/cigarette smoke

Be Strong and Say NO!

Say **NO** the very first time and every time someone offers these substances.

Remember, people who are addicted often started with **just one try**.

Choose healthy habits such as sports, hobbies and spending time with supportive friends and family.

If you or someone you know is struggling, **talk to trusted adults** like parents, teachers or school counsellors for support.

Counselling and medical help are available and important for recovery.

Science and Society: Nasha Mukh Bharat Abhiyaan

This government campaign raises awareness about **avoiding substance abuse**.

It focuses on preventing addiction, especially among children and youth.

A **National De-addiction Helpline (14446)** is available for help with drug addiction.

MULTIPLE CHOICE QUESTION

Try yourself: What is essential for growth, strength, and energy during adolescence?

A Vitamins

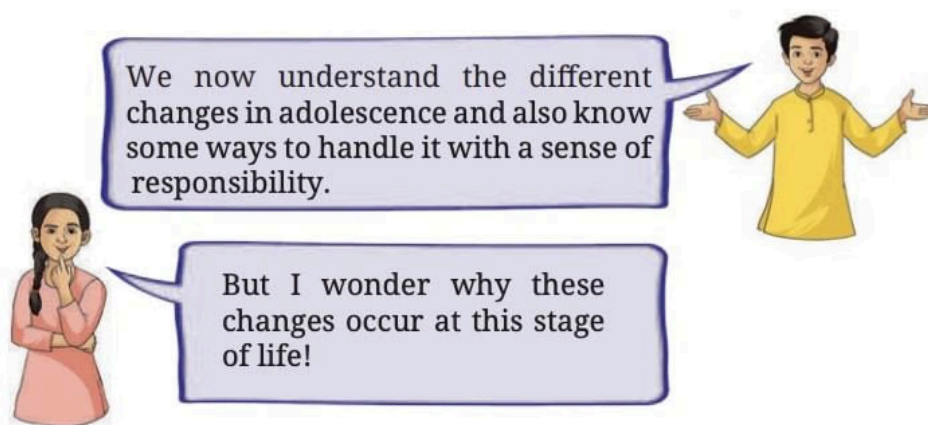
B Proteins and Carbohydrates

C Fats

D Minerals

View Solution

The 'Why' Question for Adolescence



During adolescence, the body undergoes various changes such as the onset of menstruation and other signs of puberty. These changes are primarily driven by **hormones**, which are special chemicals produced in our bodies. Hormones play a vital role in regulating different aspects of growth and development, ensuring that the body functions properly.

Hormones are made in different parts of the body and are released at the right time in response to signals from the brain. Some hormones also affect our mood and behaviour.

It is important to stay informed about these changes and seek support and guidance when needed. Making healthy decisions during this time can help build a strong foundation for the future.

Points to Remember

Adolescence is the stage from about **10 to 19 years** when a child grows into an **adult**.

Physical changes include getting **taller**, gaining **weight**, and developing features like **facial hair** in boys and **breasts** in girls.

Puberty prepares the body for having babies, and girls start their **menstrual cycle** every few weeks.

Menstruation is natural; using **sanitary pads** helps girls stay clean and healthy. Myths about periods are not true.

Teenagers may have **mood swings** and feel more **sensitive**, but these feelings can lead to **creativity** and **kindness**.

Eating **healthy foods** with protein, calcium and iron supports growth. Lack of iron or **vitamin B₁₂** can cause health problems.

Good **hygiene**, especially during periods, is important to avoid infections. Governments often provide help with sanitary products in schools.

Exercise and **sports** keep the body strong and the mind happy.

Use **social media** carefully and ask adults for help if you face problems such as **cyberbullying**.

Avoid harmful substances like **tobacco**, **alcohol** and **drugs** because they harm health and can cause **addiction**.

Hormones cause many changes in the body and feelings during adolescence.

Learning about these changes and making **good choices** helps you grow up healthy and strong.

Difficult Words and Their Meanings

Adolescence: The stage of life between childhood and adulthood (ages 10-19), marked by rapid physical, emotional and behavioural changes.

Puberty: The phase during adolescence when the body develops into an adult capable of reproduction, involving external and internal changes.

Secondary Sexual Characteristics: Physical traits (for example, facial hair in boys, breast development in girls) that distinguish males from females but are not directly involved in reproduction.

Menstrual Cycle: A natural process in girls where blood is discharged (menstruation) every few weeks, signalling reproductive health.

Menstruation: The phase of the menstrual cycle involving blood discharge, lasting 3-7 days, typically starting at puberty and ending by ages 45-55.

Hormones: Chemicals produced in the body that regulate growth, development, mood and functions such as menstruation or puberty.

Mood Swings: Sudden changes in emotions, common in adolescents, such as feeling happy one moment and low the next.

Personal Hygiene: Practices like keeping the body clean to prevent infections, especially important during adolescence.

Menstrual Hygiene: Using sanitary pads or cloth pads during menstruation to stay clean and healthy, with proper disposal to protect the environment.

Cyberbullying: Harassing others online through messages, rumours, or sharing private information; adolescents should seek help if this happens.

Substance Abuse: Regular use of harmful, addictive substances such as tobacco, alcohol, or drugs, causing health issues like lung damage or memory loss.

Addiction: A strong urge to repeatedly use harmful substances, making it hard to stop after starting.